CS60216: Safety Fundamentals of Generative AI Assignment 1

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Note: The report is concise and to the point (without AI slop)
HuggingFace ID

Models:

Reward Model
PPO - full
PPO - prefix
PPO - lora
PPO - qlora
DPO - full
DPO - prefix
DPO - lora
DPO - qlora

1 Hardware Requirements & Hyperparameters

The Jupyter (.ipynb) notebooks were executed on a Kaggle dual T4 GPU free-tier environment.

Reward Model
$epochs = 3$

PPO:
$epochs = 1,\ ppo_epochs = 2,\ kl_\beta = 0.1,\ clip_range = 0.2,\ lr = 1e-5,\ optim = AdamW$

DPO:
$epochs = 1,\ lr = 1e-5,\ optim = AdamW$

2 Results

The evaluation results are tabulated in 1

3 Comparative Study

3.1 Computation Cost

The computational time for PPO was significantly higher, taking over 5 hours to execute all strategies combined. The memory requirements were also drastically higher due to the need to keep multiple

38th Conference on Neural Information Processing Systems (NeurIPS 2024).

Figure 1: PPO Reward Progression during the fine-tuning process.

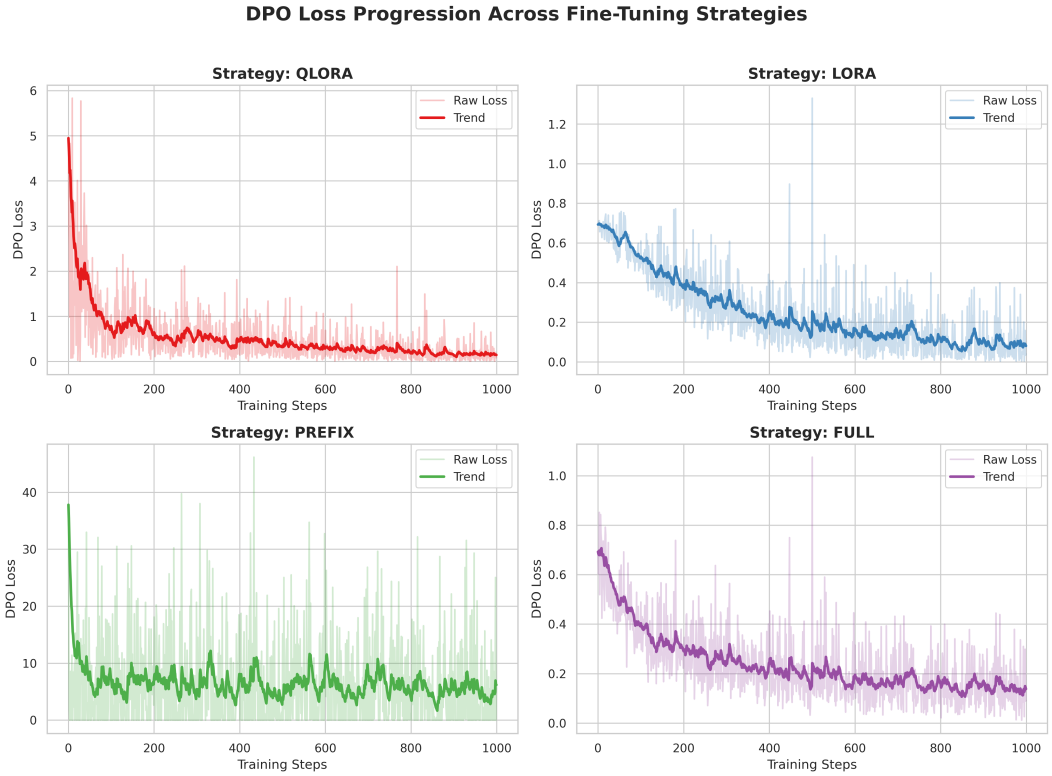

Figure 2: DPO Loss Progression split across Full, LoRA, QLoRA, and Prefix fine-tuning strategies.

Table 1: Performance comparison of PPO and DPO aligned GPT-2 models.

		BLEU		**ROUGE-L**		**Sem. Sim.**	
Method	**Variant**	More	Less	More	Less	More	Less
PPO	Full	2.3427	3.2198	0.1281	0.1403	0.5899	0.6376
	LoRA	2.4623	3.2398	0.1280	0.1403	0.5896	0.6359
	QLoRA	2.3999	3.2749	0.1282	0.1420	0.5890	0.6389
	Prefix	0.0027	0.0004	0.0690	0.0575	0.0310	0.0117
DPO	Full	2.5268	3.2488	0.1296	0.1414	0.5977	0.6423
	LoRA	2.4091	3.3078	0.1296	0.1434	0.5844	0.6346
	QLoRA	2.3976	3.2444	0.1284	0.1410	0.5828	0.6367
	Prefix	0.0008	0.0002	0.0607	0.0531	0.0658	0.0441

models (Policy, Reference, Reward, and Value) in memory simultaneously alongside the active text generation rollouts.

Conversely, DPO was highly efficient, completing all strategies in approximately 1.3 hours. As DPO only requires the Policy and Reference models and skips the active generation step during training, the overall memory footprint was much lower.

3.2 Sample Efficiency

PPO: PPO can be highly sample efficient if an incredibly accurate and robust Reward Model is available, as it can explore novel responses outside the existing dataset and learn directly from the reward signal. However, it often requires massive amounts of data to successfully train a good Reward Model first.

DPO: DPO is extremely sample efficient on existing preference datasets. Since it directly optimizes the policy based on the *more preferred* vs *less preferred* data points, it squeezes a high degree of alignment out of a smaller dataset without the intermediate bottleneck of training a reward model.

3.3 Response Quality & Stability

PPO: PPO can achieve very response quality but is unstable. It requires intense hyperparameter tuning (e.g., KL penalty beta, learning rate schedules, PPO clip range). It is also highly susceptible to *Reward Hacking*, where the model learns to exploit loopholes in the Reward Model (mode collapse).

DPO: DPO generally produces comparable or slightly superior response quality on standard benchmarks with significantly less hyperparameter tuning. It is more stable because it relies on a supervised classification loss applied to preference pairs.